

PHASE 5 – PROJECT DEVELOPMENT

1. INTRODUCTION

The Project Development phase involves the actual implementation of the **Script-Controlled ACL – Restrict Record Access Based on Field Value** project in the ServiceNow environment.

The development process includes creating users, roles, the Institution Details table, sample records, and ACLs for Read, Create, Write, and Delete operations.

2. USER CREATION

An EEE test user is created in ServiceNow through:

User Administration → Users → New

The following details are entered:

Field	Value
User ID	EEE User
First Name	EEE
Last Name	User
Email	eeeuser@gmail.com

The user is saved and used for ACL testing.

3. ROLE CREATION

Four custom roles are created through:

User Administration → Roles → New

Role	Purpose
bb1	Read access
bb2	Create access
bb3	Write access
bb4	Delete access

The required roles are assigned to the EEE test user.

4. INSTITUTION DETAILS TABLE CREATION

A custom table is created through:

System Definition → Tables → New

Label: Institution Details

Name: u_institution_details

The required fields are then added.

Field	Type
Student Roll Number	Auto Number
Student Name	Reference – User
Faculty Name	Reference – User
Branch	Choice
Email	String
Phone Number	String
Description	Multi-line String

The Branch field contains: **ECE, EEE, CSE**

5. TEST RECORD CREATION

Multiple records are created in the Institution Details table using different Branch values.

Example:

Record	Branch
Student Record 1	ECE
Student Record 2	EEE
Student Record 3	CSE

These records are required to verify the Branch-based ACL restriction.

6. READ ACL DEVELOPMENT

A Record ACL is created with the following configuration:

Property	Value
Type	Record
Operation	Read
Name	u_institution_details
Active	Yes
Advanced	Yes
Required Role	bb1
Data Condition	Branch is EEE

The advanced script is configured according to the project requirement to allow administrators and authorized users.

The Branch condition is used to restrict the applicable records to the EEE branch scenario.

7. CREATE ACL DEVELOPMENT

A Create ACL is created for:

Table: u_institution_details

Operation: Create

Required Role: bb2

This ACL controls whether an authorized user can create a new Institution Details record.

8. WRITE ACL DEVELOPMENT

A Write ACL is created for:

Table: u_institution_details

Operation: Write

Required Role: bb3

This ACL controls modification of existing records.

9. DELETE ACL DEVELOPMENT

A Delete ACL is created for:

Table: u_institution_details

Operation: Delete

Required Role: bb4

This ACL controls deletion of records.

10. ACCESS VERIFICATION DURING DEVELOPMENT

After configuring the ACLs, users are impersonated to verify the implementation.

The following access behavior is checked:

User Configuration	Expected Operation
bb1	Read permitted records
bb1 + bb2	Read + Create
bb1 + bb2 + bb3	Read + Create + Write
bb1 + bb2 + bb3 + bb4	Read + Create + Write + Delete
Administrator	Full access

11. SCREEN SHOTS

The screenshot shows the 'User' form in ServiceNow for a user named 'User - EEE User'. The form is divided into two main sections. The left section contains fields for 'User ID' (set to 'EEE User'), 'First name' (EEE), 'Last name' (User), 'Title', 'Department', 'Password needs reset', 'Locked out', 'Active' (checked), 'Identity type' (Human), and 'Internal Integration User'. The right section contains fields for 'Email' (eeeuser@gmail.com), 'Language' (None), 'Calendar integration' (Outlook), 'Time zone' (System (America/Los Angeles)), 'Date format' (System (2026-08-19)), 'Business phone', 'Mobile phone', and a 'Photo' link. Below the form are buttons for 'Update', 'Set Password', and 'Delete'. A 'Related Links' section includes links for 'View linked accounts', 'View Subscriptions', and 'Reset a password'. At the bottom, there are tabs for 'Roles (5)', 'Groups (1)', 'Delegates', and 'User Client Certificates', with a search bar and a table of roles.

FIGURE 5.1 EEE user creation.

The screenshot shows the 'Roles' list in ServiceNow. The table has columns for 'Name', 'Description', and 'Elevated privilege'. The roles listed are: bb1, bb2, bb3, bb4, bsm_legacy, bsm_legacy_admin, business_calendar_admin, business_process_admin, business_process_manager, business_process_user, business_relationship_manager, business_rule_admin, business_stakeholder, caching_admin, cannot_read_suggestions, and canvas_admin. The 'bsm_legacy_admin' role is highlighted. The table is filtered by 'Name >= bb1' and shows 1 to 20 of 700 roles.

Name	Description	Elevated privilege
bb1	Brief role purpose	false
bb2	Brief role purpose	false
bb3	Brief role purpose	false
bb4	Brief role purpose	false
bsm_legacy	Only users with the 'bsm_legacy' will see the module BSM Legacy within Dependencies View	false
bsm_legacy_admin	Only users with the 'bsm_legacy_admin' will see the module BSM Legacy within Dependencies View	false
business_calendar_admin	Access to configure and manage plugins and features related to business calendars.	false
business_process_admin		false
business_process_manager		false
business_process_user		false
business_relationship_manager	Person(s) responsible for both business strategy and business value in an effort to advance business objectives.	false
business_rule_admin	Can manage Business Rules	false
business_stakeholder	Use the Business Stakeholder role to provide authorized users to approve, view/read records across the organization and view reports. Note there is a fee associated with this role. Do not assign it to users without confirming your organization has the appropriate entitlement. Please review the documentation for additional details.	false
caching_admin	Role for monitoring and configuring caching framework related tables	false
cannot_read_suggestions	Users with this role are not authorized to use suggestion service.	false
canvas_admin	Admin role for Canvas Core Application	false

FIGURE 5.2 Custom role creation.

servicenow All Favorites History Workspaces Admin User - EEE User Search

User EEE User Update Set Password Delete

Locked out ☐ Mobile phone

Active ☒ Photo Click to add...

Identity type Human

Internal Integration User ☐

Update Set Password Delete

Related Links

[View linked accounts](#)

[View Subscriptions](#)

[Reset a password](#)

Roles (5) Groups (1) Delegates User Client Certificates

Role Search Actions on selected rows... Edit...

User = EEE User

Role	State	Inherited	Inheritance Count
bb1	Active	false	
bb2	Active	false	
bb3	Active	false	
snc_required_script_writer_permission	Active	true	1
bb4	Active	false	

1 to 5 of 5

FIGURE 5.3 Role assignment.

Table Institution Details Delete Update Delete All Records

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label Institution Details Application Global

* Name u_institution_details Remote Table

Columns Controls Application Access

Table Columns for text Search 1 to 13 of 13 New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)		32	false
Created by	String	(empty)		40	false
Branch	Choice	(empty)		40	false
Student Name	Reference	User		32	false
Created	Date/Time	(empty)		40	false
Faculty Name	Reference	User		32	false
Description	String	(empty)		40	false
Updates	Integer	(empty)		40	false
Student Roll Number	String	(empty)		40	false

FIGURE 5.4 Institution Details table.

servicenow All Favorites History Workspaces Admin Table - Institution Details

Search

Table Institution Details

Columns Controls Application Access

Table Columns for text Search 1 to 13 of 13 New

Dictionary Entries

Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)	32		false
Created by	String	(empty)	40		false
Branch	Choice	(empty)	40		false
Student Name	Reference	User	32		false
Created	Date/Time	(empty)	40		false
Faculty Name	Reference	User	32		false
Description	String	(empty)	40		false
Updates	Integer	(empty)	40		false
Student Roll Number	String	(empty)	40		false
Updated by	String	(empty)	40		false
Phone Number	String	(empty)	40		false
Email	String	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Insert a new row...					

Show desktop

FIGURE 5.5 Table fields.

servicenow All Favorites History Workspaces Admin Institution Details

Search

Institution Details Branch Search

Actions on selected rows... New

All

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
CSE			Abraham Lincoln		Adela Cervantsz	IN0003
(empty)			(empty)		(empty)	
ECE			Abel Tuter		Adela Cervantsz	IN00001
EEE			Abraham Lincoln		System Administrator	IN0002

1 to 4 of 4

FIGURE 5.6 Sample records.

servicenow All Favorites History Workspaces Admin Access Control - New Record

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record Application: Global

* Operation: read Active: ☒

Admin overrides: ☒ Advanced: ☒

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details] -- None --

Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More info](#)

Requires role

Role
+

Insert a new row...

FIGURE 5.7 Read ACL.

servicenow All Favorites History Workspaces Admin Access Control - New Record

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record Application: Global

* Operation: create Active: ☒

Admin overrides: ☒ Advanced: ☐

Decision Type: Allow If

Protection policy: -- None -- Tables

* Name: Institution Details [u_institution_details] -- None --

Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More info](#)

Requires role

Role
+

Insert a new row...

FIGURE 5.8 Create ACL.

servicenow All Favorites History Workspaces Admin **Access Control - New Record** Search

Access Control New record Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record Application: Global

* Operation: write Active: ☒

Admin overrides: ☒ Advanced: ☐

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details] -- None --

Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
Insert a new row...

FIGURE 5.9 Write ACL.

servicenow All Favorites History Workspaces Admin **Access Control - New Record** Search

Access Control New record Submit

Warning: A role, security attribute, data condition, script or ACL control via reference fields is required to properly secure access with this ACL.

* Type: record Application: Global

* Operation: delete Active: ☒

Admin overrides: ☒ Advanced: ☐

Decision Type: Allow If

Protection policy: -- None --

* Name: Institution Details [u_institution_details] -- None --

Description:

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
Insert a new row...

FIGURE 5.10 Delete ACL.

The screenshot shows the ServiceNow 'Institution Details' page. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Process Mining Workspace', and a search bar. Below the navigation bar, there's a filter section with 'Institution Details' selected and a search input. The main table has columns: Branch, Description, Email, Faculty Name, Phone Number, Student Name, and Student Roll Number. Only one record is visible for Branch 'EEE'.

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
EEE			Abraham Lincoln		System Administrator	IN0002

At the bottom, there is a pagination control showing '1 to 1 of 1'.

FIGURE 5.11 User access verification.

The screenshot shows the ServiceNow 'Institution Details' page with a list of four records. The first record is for Branch 'CSE' with Faculty Name 'Abraham Lincoln' and Student Name 'Adela Cervantsz'. The second record is for Branch '(empty)' with Faculty Name '(empty)' and Student Name '(empty)'. The third record is for Branch 'ECE' with Faculty Name 'Abel Tuter' and Student Name 'Adela Cervantsz'. The fourth record is for Branch 'EEE' with Faculty Name 'Abraham Lincoln' and Student Name 'System Administrator'.

Branch	Description	Email	Faculty Name	Phone Number	Student Name	Student Roll Number
CSE			Abraham Lincoln		Adela Cervantsz	IN0003
(empty)			(empty)		(empty)	
ECE			Abel Tuter		Adela Cervantsz	IN0001
EEE			Abraham Lincoln		System Administrator	IN0002

At the bottom, there is a pagination control showing '1 to 4 of 4'.

FIGURE 5.12 Administrator access verification.

12. DEVELOPMENT OUTCOME

At the end of the development phase, the ServiceNow environment should contain:

- Configured test user.
- Four custom roles.
- Institution Details table.
- Required fields and sample records.
- Four ACL configurations.
- Branch-based Read restriction.
- Controlled Create, Write, and Delete operations.

The completed configuration is then ready for detailed **Project Testing** in the next phase.